Relation Changes Matter: Cross-Temporal Difference Transformer for Change Detection in Remote Sensing Images

Kai Zhang, *Member, IEEE*, Xue Zhao, Feng Zhang, Lei Ding, Jiande Sun, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*— Thanks to their capability of modeling global information, transformers have been recently applied to change detection (CD) in remote sensing images. Generally, the changes in terms of shape and appearance of objects lead to relation changes among these objects in multitemporal images. However, in this context, the attention mechanism in transformers has not been fully explored yet to learn relation changes in the observed scenes. In this article, we analyze the relation changes in multitemporal images and propose a cross-temporal difference (CTD) attention to capturing these changes efficiently. Through the CTD attention, the changed areas are distinguished better from the unchanged areas. Based on the CTD attention, two CTD-transformer encoders are constructed to extract the features of changed areas from the embedded tokens of multitemporal images in a cross manner. Then, the extracted features at the coarse scale are further improved to the fine-scale by the corresponding CTD-transformer decoders. In addition, consistency-perception blocks (CPBs) are designed to preserve the structures and contours of changed areas. Finally, all extracted features from multitemporal images are concatenated to produce the desired change map. Compared to state-of-the-art methods, experimental results on LEVIR-CD, WHU-CD, and CLCD datasets demonstrate that the proposed method produces better performance. The source code is available at https://github.com/RSMagneto/CTD-Former.**

***Index Terms*— Change detection (CD), cross-temporal difference (CTD) attention, relation changes, remote sensing, transformer.**

I. Introduction

CHANGE detection (CD) aims to infer the changes in a given observed scene from two remote sensing images collected at different times [1]. It has become a popular topic in the remote sensing community and has been extensively applied to many related fields, including environmental monitoring [2], disaster evaluation [3], and urban planning [4]. In this context, obtaining high-precision CD maps plays an increasingly important role in the interpretation of observed scenes and has crucial importance to protect natural resources.

With high-resolution (HR) remote sensing images accumulated constantly by many satellites, such as QuickBird, IKONOS, and WorldView, the evolution of features of objects on the ground becomes possible. While multitemporal HR remote sensing images provide fine-grained structures of objects, they also contain more complicated change types, which bring new challenges to CD. In recent years, the performance boost of CD has benefited from the powerful representation capacities of deep neural networks (DNNs). Many DNNs have been exploited to extract the change information from the multitemporal remote sensing images [5]. As one of the most popular architectures, convolution neural networks (CNNs) have gained great attention. Many CNN-based CD methods [6], [7], [8] have been proposed to produce good results. However, these methods have difficulties in preserving the structural integrity of large-scale changed areas owing to the locality of convolution in CNN. To expand the receptive field of the convolution operation, pyramid structures [9], [10] are considered to extract the features of images at different scales. Besides, hybrid CNNs [11], [12] combine conventional convolutions and dilated convolutions to extract the local textures and nonlocal semantic relationships in multitemporal remote sensing images simultaneously.

Recently, the proposed transformer has shown great potential for modeling global dependencies in images and has been applied to various tasks, such as objection detection [13] and pan-sharpening [14], [15]. Owing to the excellent performance, many transformer-based methods [16], [17], [18], [19], [20], [21] have been proposed to improve CD performance. However, these methods only used the transformer as a straightforward feature extractor. The attention mechanism of transformers for the CD task is far from being fully explored and exploited. According to [22] and [23], the self-attention in transformers can be described by a directed graph, where each token is regarded as a vertex and the inner product between two tokens represents an edge. Therefore, the self-attention in transformers records the similarity relations among tokens in images apart from feature extraction. Changes in

Manuscript received 7 December 2022; revised 31 March 2023 and 11 May 2023; accepted 29 May 2023. Date of publication 31 May 2023; date of current version 13 June 2023. This work was supported in part by the Natural Science Foundation of China under Grant 61901246, in part by the China Postdoctoral Science Foundation under Grant 2019TQ0190 and Grant 2019M662432, in part by the Scientific Research Leader Studio of Jinan under Grant 2021GXRC081, and in part by the Joint Project for Smart Computing of Shandong Natural Science Foundation under Grant ZR2020LZH015. *(Corresponding author: Kai Zhang.)*

Kai Zhang, Xue Zhao, Feng Zhang, and Jiande Sun are with the School of Information Science and Engineering, Shandong Normal University, Jinan 250358, China (e-mail: zhangkainuc@163.com; meiyouzidexigua@163.com; fengzhangpl@163.com; jiandesun@hotmail.com).

Lei Ding is with the Computer Science Department, PLA Strategic Support Force Information Engineering University, Zhengzhou 450001, China (e-mail: dinglei14@outlook.com).

Lorenzo Bruzzone is with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: lorenzo.bruzzone@unitn.it).

Digital Object Identifier 10.1109/TGRS.2023.3281711

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

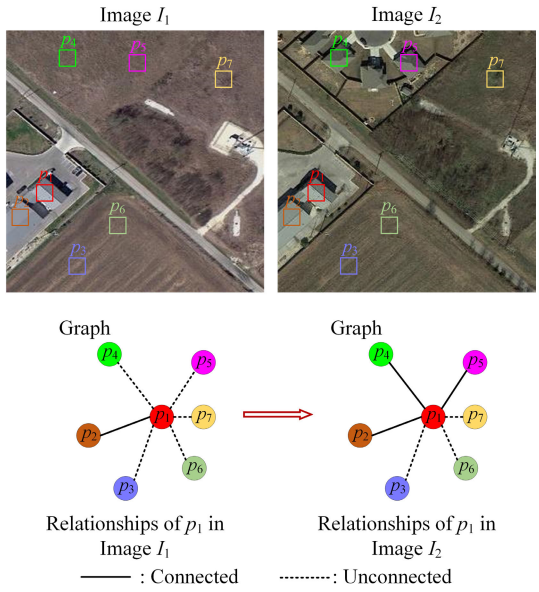


Fig. 1. Relation changes in multitemporal images.

multitemporal remote sensing images mean not only the changes in shape and appearance of objects but also the changes in terms of the similarity relations among objects. The aforementioned methods in [16], [17], [18], [19], [20], and [21] utilize the attention mechanism to model the relations among tokens from single-temporal or multitemporal features, but the relation changes are not sufficiently mined to emphasize the features of changed areas. Thus, the features of changed and unchanged areas are treated without distinction, which limits the CD performance of these methods.

Fig. 1 demonstrates the relation changes in multitemporal images. For patch p_1 in Image I_1, we can see that it is similar to patch p_2 and not similar to other patches, such as p_4 and p_5. However, patches p_4 and p_5 become buildings and are more similar to patch p_1 in Image I_2. Thus, the patches p_4 and p_5 are connected to patch p_1 in the constructed graph of Image I_2. The second row in Fig. 1 shows the connection changes between the two graphs from multitemporal images. The evolution in terms of connections means that the similarity relations of patches are changed. So, relation changes also reflect some extent the changes in objects on the ground. The extraction of relation changes will be helpful to model the spatial information more comprehensively and infer the changed areas more efficiently.

To model the relation changes in multitemporal images, we propose a novel cross-temporal difference (CTD) attention, which is embedded into the transformer to form the CTD-Former for efficient extraction of the changed features. Specifically, we first revisit the self-attention in transformers and deeply analyze the relation changes among objects in multitemporal images. Then, the CTD attention is derived by calculating the differences between the similarity matrices of multitemporal images. According to the CTD attention, CTD-transformer encoders are designed to impose the relation changes on the tokens of multitemporal images. Next, the outputs of the encoders are fed into CTD-transformer decoders to enhance their details in them. Besides, we adopt consistency-perception blocks (CPBs) to further enrich the features of CD, in which the dilated convolution and deformable convolution are introduced to better enlarge the receptive field and match the contours of changed areas, respectively. Finally, we integrate the features extracted from two images and estimate the change map effectively. We conduct extensive experiments on LEVIR-CD, WHU-CD, and CLCD datasets, which are composed of bi-temporal mutispectral images. The proposed CTD-Former shows better results in terms of efficiency and accuracy compared to state-of-the-art techniques. Ablation studies illustrate that the proposed CTD attention distinguishes the changed features more efficiently.

The contributions of the article are summarized as follows.

1) We propose a new attention mechanism, CTD attention, by deeply excavating the similarity relation changes between multitemporal images. The CTD attention is derived from the differences between similarity matrices of multitemporal images.
2) We construct CTD-transformer encoders according to the CTD attention. In combination with the differences in terms of relations, the proposed CTD-transformer encoders highlight the features of changed areas efficiently.
3) We design CPBs to preserve the spatial structures of the changed areas, which are equipped with dilated and deformable convolution layers. They ensure the consistency of changed areas existing within several adjacent tokens.

The rest of the article is organized as follows. Section II introduces some methods based on traditional paradigms and DNNs. Section III describes the proposed CTD-Former in detail. Section IV demonstrates comparison results and ablation studies on LEVIR-CD, WHU-CD, and CLCD datasets. Finally, Section V gives the conclusion.

II. Related Work

A. Traditional CD Methods

During the last decades, CD performance has achieved a large improvement and many traditional methods have been proposed. For example, some transforms, such as principal component analysis (PCA) [24] and independent component analysis (ICA) [25], were used to infer the change information between multitemporal remote sensing images. Nielsen et al. [26] and Nielsen [27] proposed multivariate alteration detection (MAD), which was derived from canonical correlation analysis (CCA). Besides, change vector analysis (CVA) [28] and its variants [29] were proposed to analyze the differences between changed and unchanged information in the polar domain. These methods are easy to implement and use. However, they only focus on the modeling of the spectral information and ignore the influences of the spatial information.

Subsequently, machine learning-based methods were proposed to integrate spatial and spectral information in images. For instance, Lv et al. [30] designed a hybrid conditional

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

random field model to extract informative spatial and spectral features. Habib et al. [31] employed the support vector machine (SVM) to identify the changed and unchanged pixels in multitemporal remote sensing images. Markov random fields were also considered in [32] to fuse the spectral features and spatial content. In addition, various priors of changed information, such as sparsity [33], low rank [34], and multiscale property [35], in the observed scenes were also explored for CD. For these methods, the hand-craft features limited their performance because it is difficult for them to represent the changed information in complicated scenes.

B. DNN-Based CD Methods

Nowadays, DNNs have evolved to become the main techniques for CD because they can produce accurate CD results. Compared to traditional methods, DNN-based methods can hierarchically extract the spatial, spectral, and temporal features of the multitemporal images in feature space. The learned features by DNNs have more powerful representation capacities than the hand-crafted features from traditional methods. So, DNN-based methods achieve higher CD precision and the introductions of DNNs also bring an opportunity of performance improvement for traditional CD methods. Saha et al. [36] used CVA to analyze the differences between the features extracted by DNNs. As a commonly-used network, CNN is often adopted to identify the changed information in multitemporal remote sensing images. For example, Daudt et al. [6] constructed three fully convolution (FC) networks for CD, including FC-early fusion (FC-EF), FC-Siamese-concatenation (FC-Siam-conc), and FC-Siamese-difference (FC-Siam-diff). Shi et al. [7] introduced deep supervision to constrain the intermediate feature maps, which were extracted by two CNNs. Chen et al. [8] also adopted a Siamese CNN to learn the changed information from augmented samples. Liu et al. [37] introduced depth-wise separable convolutions into the original FC network to reduce its model size, which made the proposed network lighter. Similarly, Han et al. [38] also proposed a lightweight FC network for CD to improve its flexibility.

As the number of network architectures increases, researchers in this field consider other DNNs for CD. For instance, some researchers adopted recurrent neural networks (RNNs) to learn the temporal dependency between multitemporal remote sensing images. Mou et al. [11] combined a CNN and an RNN to extract the spatial-spectral-temporal features jointly. Chen et al. [39] utilized a stacked RNN to project the extracted features into a new space. Besides, the generative adversarial network (GAN) was used in [40] to synthesize cross-temporal images, by which the changed areas were found easily. Jiang et al. [41] proposed a semi-supervised GAN to use both labeled and unlabeled samples in images. Further, Ren et al. [42] advanced an unsupervised CD method based on GAN, in which GAN was employed to produce better-registered images. In addition, the graph convolutional network (GCN) was also exploited to make full use of the labeled and unlabeled samples. In [43], the graph was constructed by superpixel segmentation and the homogeneity of changed information was modeled in GCN. Tang et al. [44] designed a multiscale dynamic GCN to capture the local and nonlocal features in images.

Recently, researchers have focused more on the attention mechanism. Ding et al. [45] achieved semantic reasoning by the spatial consistency and temporal correlation calculated by self-attention. Peng et al. [46] used spatial attention and channel attention to extract the content information from spatial and channel dimensions, respectively. Song and Jiang [47] proposed similar attention modules to guide the learning of the change information. Huang et al. [48] applied the attention mechanism to the fusion of the features from the sub-networks of multitemporal remote sensing images. Very recently, transformers with the self-attention mechanism have obtained state-of-the-art results in various language and vision tasks [49], [50]. Inspired by its success, the transformer was also extended to CD [16], [17], [18], [19], [20], [21]. For example, Bandara and Patel [21] presented an encoder consisting of stacked transformer blocks as the feature extractor. The aforementioned CD methods using transformers exploit an architecture that is the combination of CNN and transformer, which extract the local and global features simultaneously. Zhang et al. [51] established a pure transformer network that was exclusively composed of Swin transformer blocks [52].

Inspired by the strong representation capabilities of transformers in the bitemporal image transformer (BIT) [17], we propose CTD-Former aims to characterize the attention differences between multitemporal images and utilize the relation changes in the cross-temporal attention differences to adaptively enhance the changed features. Compared to BIT, we propose CTD attention by inferring the relation changes between multitemporal images. BIT directly concatenates the features from different images and only uses the vanilla self-attention in transformers to learn the global information in images. Besides, CPBs are introduced into CTD-Former to extract the information at larger scales and preserve the shape information of changed areas better via deformable convolution.

III. Proposed Method

In this section, we first introduce the overall framework of the proposed CTD-Former, followed by the CTD-Transformer, CPB, and loss function. Finally, the proposed CTD-Former is trained to detect the changed areas in multitemporal images.

A. Overall Architecture

The framework of the proposed CTD-Former is shown in Fig. 2 In CTD-Former, the features $E_1 \in \mathbb{R}^{(H/2)\times(W/2)\times C}$ and $E_2 \in \mathbb{R}^{(H/2)\times(W/2)\times C}$ are first extracted by the corresponding encoders from multitemporal images $I_1 \in \mathbb{R}^{H\times W\times B}$ and $I_2 \in \mathbb{R}^{H\times W\times B}$. H and W are the height and width of the input images, respectively. B is the number of the bands in I_1 and I_2. The number of channels in E_1 and E_2 is denoted by C. In the proposed CTD-Former, a modified ResNet-18 [53] is employed as the encoder for feature extraction. Specifically, we remove the max-pooling layer in ResNet-18 and set the convolution stride as 1 in the original last three strided

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

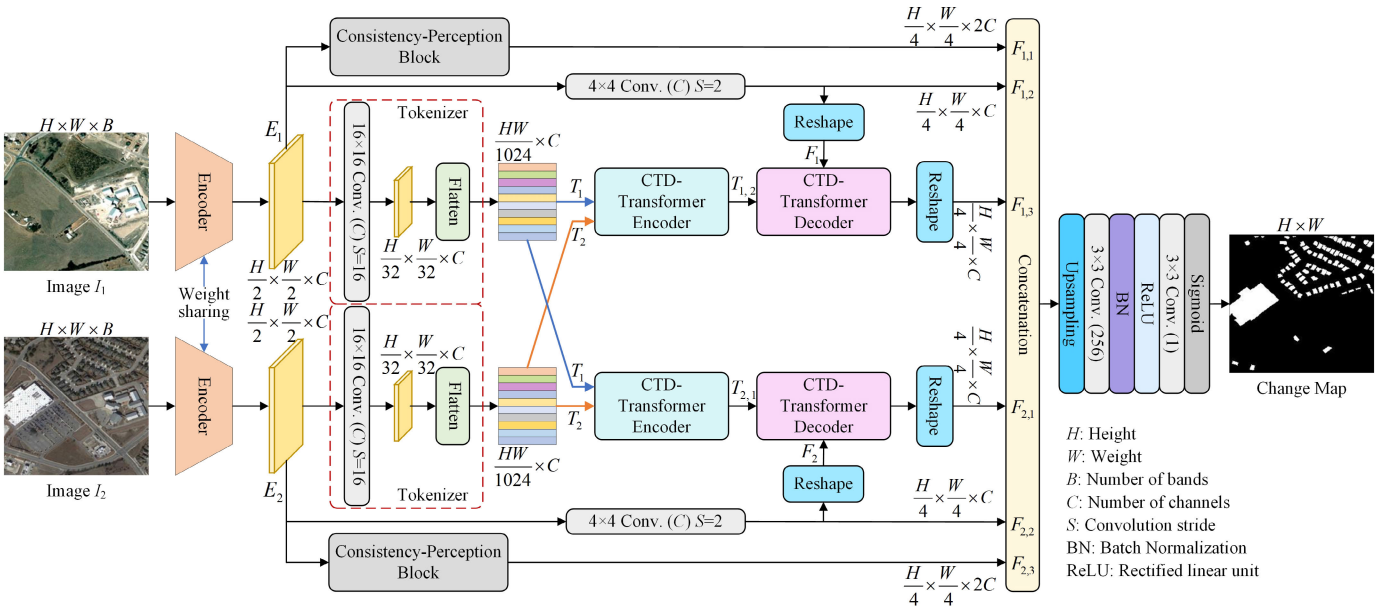


Fig. 2. CTD-Former architecture.

convolution layers. A point-wise convolution layer is introduced to condense the features into C channels. Next, E_1 and E_2 are tokenized by a strided convolution layer and a flattened operation to obtain $T_1 \in \mathbb{R}^{(\mathrm{HW}/1024)\times C}$ and $T_2 \in \mathbb{R}^{(\mathrm{HW}/1024)\times C}$, where each row stands for a token. Typically, the filter size and stride in the tokenizer of Fig. 2 are set as 16×16 and 16 for demonstration purposes, whose influences will be analyzed in Section IV-G. The tokenization in Fig. 2 can be regarded as a mapping, which projects the nonoverlapping 16×16 patches in E_1 or E_2 as row vectors in T_1 or T_2. Then, T_1 and T_2 are fed into the two CTD-transformer encoders in a cross manner to capture the relation changes in multitemporal images.

To highlight the features of changed areas, the estimated relation changes are combined with E_1 and E_2 in the following CTD-transformer decoders. E_1 and E_2 are first convolved by 4×4 filters with stride 2 and then reshaped as $F_1 \in \mathbb{R}^{(\mathrm{HW}/16)\times C}$ and $F_2 \in \mathbb{R}^{(\mathrm{HW}/16)\times C}$, which are fed into the CTD-transformer decoders. In addition, CPBs are introduced to preserve the contours and edges of changed areas. Finally, all features are concatenated to infer the change map of the observed scene.

B. CTD-Transformer

1) Revisiting the Self-Attention in Transformer: Let $X \in \mathbb{R}^{n\times c}$ be an embedded feature, where n is the number of tokens and c indicates the dimension of tokens. According to the self-attention mechanism, the output $\tilde{x}_i$ of the ith token x_i in X is calculated as

$$\tilde{x}_i = \frac{1}{Z_i(X)} \sum_{j=1}^{n} g\left(x_i \cdot W_Q, x_j \cdot W_K\right) x_j \cdot W_V \tag{1}$$

where W_Q, W_K, and W_V are linear mappings. $\cdot$ is the matrix multiplication. $g(\cdot,\cdot)$ is a pairwise function computing similarity relation between two tokens x_i and x_j. $Z_i(X)$ is the corresponding normalization factor of the ith token x_i. Regarding each token as a vertex, a fully-connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ can be constructed, where $\mathcal{V}$ is the set of all tokens in X and the edge $(x_i, x_j) \in \mathcal{E}$ records the relationship between the two vertexes. Specifically, when $e^{x_i W_Q W_K^T x_j^T}$ is employed as $g(\cdot)$, the self-attention mechanism for all tokens is defined as

$$Y = A \cdot V,\ A = \mathrm{softmax}\left(\frac{Q \cdot K^T}{\sqrt{s}}\right) \tag{2}$$

where the query, key, and value are $Q = X \cdot W_Q$, $K = X \cdot W_K$, and $V = X \cdot W_V$. s is a scale factor. Thus, A can be thought of as an adjacency matrix, which contains the similarity relations between two tokens.

2) Modeling the Relation Changes by CTD Attention: According to the tokenization in Fig. 2, the tokens in T_1 and T_2 are the projected features of the nonoverlapping 16×16 patches in E_1 and E_2. So, each token represents a corresponding patch in E_1 and E_2. Then, the relations among these patches can be captured by computing the attention matrix in (2) from the corresponding tokens in T_1. Take the ith token t_1^i in T_1 as an example, the relations with other tokens are calculated as

$$a_1^i = \mathrm{softmax}\left(\frac{q_1^i \cdot K_1^T}{\sqrt{s}}\right) \tag{3}$$

where q_1^i is the ith row of Q_1. Q_1 and K_1 are the query and key of T_1 after linear projections, respectively. a_1^i is a vector, which represents the relations between t_1^i with other tokens in T_1. However, only estimating the self-attention among patches from T_1 cannot model the changed relations in multitemporal images, because T_1 is inferred from the single-temporal image.

To describe the changed relations, we calculate the similarity relations between t_1^i and the tokens from T_2, which are then subtracted by a_1^i to obtain the relation changes. This process

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

is written as

$$m_1^i = \text{softmax}\left(\frac{(d_1^i)^2}{\sqrt{s}}\right)$$
$$d_1^i = q_1^i \cdot K_1^T - q_1^i \cdot P_2^T \tag{4}$$

where P_2 is the linear projection of T_2. $(\cdot)^2$ denotes the element-wise square.

If there are no changes in the corresponding patch of the jth token t_2^j in T_2, the inner product between q_1^i and k_1^j will tend to be the same as that between q_1^i and p_2^j. So, $d_1^i(j)$ will be close to zero. k_1^j and p_2^j are the jth rows in K_1 and P_2, and they also correspond to the projections of the jth patches in E_1 and E_2, respectively. $d_1^i(j)$ is the jth element of the difference vector d_1^i. If the jth patch in image I_2 is changed, the inner products in (4) will be very different. For the vanilla attention in (2), the query q is used to find all similar tokens from K, which simultaneously contain changed and unchanged tokens. Thus, the information in changed and unchanged areas is emphasized by this attention. The unchanged areas cannot be suppressed efficiently. For the proposed CTD attention, the subtraction operation between $q_1^i K_1^T$ and $q_1^i P_2^T$ not only enhances the information of changed areas but also suppresses the influences of unchanged areas. When the jth patch in image I_2 becomes more similar to the ith patch in image I_1, the inner product between q_1^i and p_2^j will be larger. Then, the difference $d_1^i(j)$ will be negative. Conversely, the difference $d_1^i(j)$ will be positive, if the jth patch in image I_2 turns into the one that is not similar to the ith patch in image I_1. Considering the signs of elements in d_1^i, the element-wise square is imposed on them. Then, the changes in multitemporal images are reflected by $(d_1^i(j))^2$. If the jth patch in image I_2 is changed, $(d_1^i(j))^2$ will be larger than zero. Otherwise, $(d_1^i(j))^2$ will be close to zero. Finally, the attention score is normalized by the softmax function.

Considering all tokens in T_1, the relation changes are formulated as

$$M_1 = \text{softmax}\left(\frac{\left(Q_1 \cdot K_1^T - Q_1 \cdot P_2^T\right)^2}{\sqrt{s}}\right). \tag{5}$$

Then, the CTD attention for image I_1 can be expressed as

$$A_1 = M_1 \cdot V_1 \tag{6}$$

where V_1 is the projected value of T_1.

Similarly, the CTD attention for image I_2 is computed as

$$A_2 = \text{softmax}\left(\frac{\left(Q_2 \cdot K_2^T - Q_2 \cdot P_1^T\right)^2}{\sqrt{s}}\right) V_2 \tag{7}$$

where Q_2, K_2, and V_2 are the query, key, and value of T_2, respectively. P_1 is the linear projection of T_1 in the CTD attention of image I_2. Through the CTD attention, the features of changed areas are highlighted and the unchanged areas are suppressed.

3) Toy Example: To explain the effectiveness of the proposed CTD attention, we demonstrate it on a toy example in Fig. 3. In the toy example, I_1 and I_2 are composed of five classes of patches and the changed patches in I_2 are outlined by red rectangles. The size of each patch is 5×5 and each image is segmented into 64 patches without overlap. To make this case easier to explain, we omit linear projections. Then, the pixel values in each patch are vectorized as the corresponding token. So, each image is tokenized as a matrix with the size of 64×75. The number of tokens is 64 and the dimension of each token is 75. We directly regard the token as its query, key, and value.

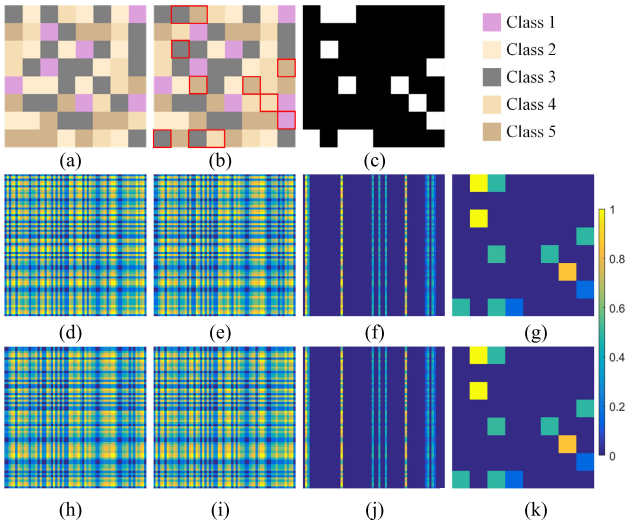


Fig. 3. Toy example. (a)–(c) Bi-temporal images (images I_1 and I_2) and the corresponding ground truth. (d) and (e) Similarity relations in (5), i.e., $Q_1 K_1^T$ and $Q_1 P_2^T$. (f) Differences between (d) and (e), i.e., $(Q_1 K_1^T - Q_1 P_2^T)^2$. (g) Changed feature, i.e., reshape(A_1). (h) and (i) Similarity relations in (7), i.e., $Q_2 K_2^T$ and $Q_2 P_1^T$. (j) Differences between (h) and (i), i.e., $(Q_2 K_2^T - Q_2 P_1^T)^2$. (k) Changed feature, i.e., reshape(A_2).

According to (5), the similarity relations are calculated as $Q_1 \cdot K_1^T$ and $Q_1 \cdot P_2^T$, shown in Fig. 3(d) and (e). Through the subtraction between $Q_1 \cdot K_1^T$ and $Q_1 \cdot P_2^T$, the relation changes can be extracted. From the relation changes in Fig. 3(f), we can see that most of the similar relations keep unchanged because only a small number of patches are changed in I_2. Finally, the relation changes are multiplied by the value V_1 of I_1 to obtain A_1. We reshape A_1 to the size of I_1 and show it in Fig. 3(g). It can be seen that the locations of the extracted features correspond to the changed areas in the ground truth. The pixel values of the patches in Fig. 3(g) reflect the degree of changes between I_1 and I_2. In addition, we also compute the CTD attention in (7) and show the results in Fig. 3(h)–(k). Similar results can be also derived from (7). Therefore, the proposed CTD attention can efficiently extract the features of changed areas because the relationship changes between I_1 and I_2 are modeled adequately.

4) CTD Transformer Encoder: According to the CTD attention in (5) and (7), we design CTD transformer encoders to further extracted the features of changed areas from images I_1 and I_2. Fig. 4 demonstrates the architecture of the CTD transformer encoder for the image I_1. The encoder in Fig. 4 is composed of a multihead CTD attention block and a multilayer perceptron (MLP) block. Then, the process in Fig. 4 can be formulated as

$$z_1 = \text{MCTDAtt}(\text{LN}(T_1), \text{LN}(T_2)) + T_1$$
$$T_{1,2} = \text{MLP}(\text{LN}(z_1)) + z_1 \tag{8}$$

where $\text{LN}(\cdot)$ denotes the layer normalization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

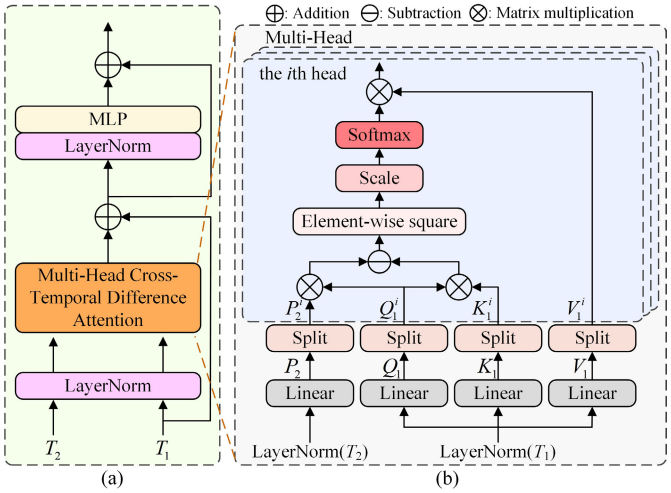


Fig. 4. Architecture of (a) CTD transformer encoder and (b) multihead CTD attention.

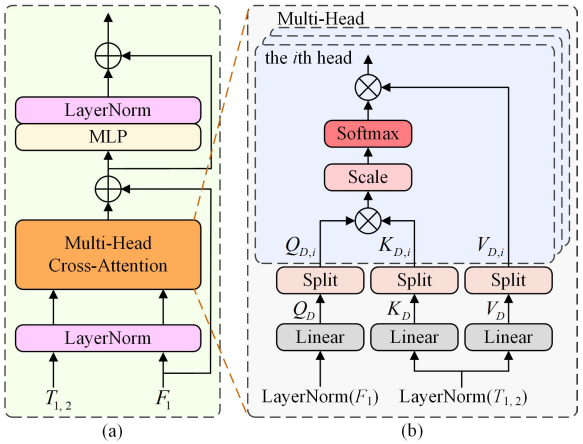


Fig. 5. Architecture of (a) CTD transformer decoder and (b) multihead cross-attention.

In (8), the multihead mechanism is considered to exploit the representation in different subspaces. When multihead is used, more attention patterns among query, key, and value can be explored efficiently. The feature dimension is projected from C to b. So, the sizes of Q_1, K_1, and V_1 are $(\mathrm{HW}/1024) \times b$. P_2 also has the same size. Here, b equals 512. To reduce the computational complexity, these projected features are partitioned into N parts along the feature dimension to produce Q_1^i, K_1^i, V_1^i, and P_2^i. So, their size is $(\mathrm{HW}/1024) \times (d/N)$. N stands for the number of heads in Fig. 4(b), which is specifically set as 8. Finally, the multihead CTD attention MCTDAtt$(\cdot)$ is written as

$$\mathrm{MCTDAtt}(\mathrm{LN}(T_1), \mathrm{LN}(T_2)) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_N) W_O \tag{9}$$

where $\mathrm{head}_i = \mathrm{CTDAtt}(Q_1^i, K_1^i, V_1^i, P_2^i)$, CTDAtt$(\cdot)$ indicates the attention in (6). Then, the outputs of all heads are concatenated and mapped by the learnable projection matrix W_O.

The inputs of the CTD transformer encoder of the image I_2 are switched and the same structure as that in Fig. 4 is adopted. The encoder for I_2 is

$$\begin{aligned} z_2 &= \mathrm{MCTDAtt}(\mathrm{LN}(T_2), \mathrm{LN}(T_1)) + T_2 \\ T_{2,1} &= \mathrm{MLP}(\mathrm{LN}(z_2)) + z_2. \end{aligned} \tag{10}$$

5) CTD Transformer Decoder: Through the encoder in Fig. 4, the relation changes are embedded in $T_{1,2}$ and $T_{2,1}$. However, the features E_1 and E_2 are down-sampled to the coarse scale by the corresponding tokenizers for the reduction of the computational complexity in the CTD transformer encoder. So, the features in $T_{1,2}$ and $T_{2,1}$ are limited by the coarse scale. To pass the relation changes in $T_{1,2}$ and $T_{2,1}$ to the fine scale, CTD transformer decoders are constructed to combine E_1 and E_2. Fig. 5 shows the architecture of the CTD transformer decoder for the image I_1. Before sending E_1 into the decoder, it is convoluted by a strided convolution layer and then reshaped as F_1 in a channel-wise manner. Then, the query $Q_D \in \mathbb{R}^{(\mathrm{HW}/16)\times l}$ is obtained by mapping F_1. $T_{1,2}$ is projected by linear projections to obtain the key $K_D \in \mathbb{R}^{(\mathrm{HW}/1024)\times l}$ and value $V_D \in \mathbb{R}^{(\mathrm{HW}/1024)\times l}$. l is the feature dimension, which is set as 512 in this article. In the multihead cross-attention of Fig. 5, Q_D, K_D, and V_D are split into R parts along the horizontal dimension to reduce the complexity. $R = 8$ is the number of heads. Then, the inputs of the ith head are $Q_{D,i} \in \mathbb{R}^{(\mathrm{HW}/16)\times(l/R)}$, $K_{D,i} \in \mathbb{R}^{(\mathrm{HW}/16)\times(l/R)}$, and $V_{D,i} \in \mathbb{R}^{(\mathrm{HW}/16)\times(l/R)}$. So, the cross-attention among them is defined as

$$\mathrm{CAtt}\left(Q_{D,i}, K_{D,i}, V_{D,i}\right) = \mathrm{softmax}\left(\frac{Q_{D,i} \cdot K_{D,i}^T}{\sqrt{s}}\right) V_{D,i}. \tag{11}$$

In the multihead cross attention, the outputs of all heads are integrated, which can be written as

$$\mathrm{MCAtt}\left(\mathrm{LN}\left(T_{1,2}\right), \mathrm{LN}(F_1)\right) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_N) W_D \tag{12}$$

where $\mathrm{head}_i = \mathrm{CAtt}(Q_{D,i}, K_{D,i}, V_{D,i})$ and $W_D \in \mathbb{R}^{l\times C}$ is a projection matrix. Finally, we can obtain a refined feature at fine scales, which is reshaped and concatenated with other features. In the same way, F_2 is also combined with the CTD transformer decoder to refine the changed feature in image I_2.

C. Consistency-Perception Block

In Fig. 2, the tokenization is equivalent to project the nonoverlapping patches with the size of 16×16 in E_1 or E_2 as a vector. Although the relations between patches are depicted by the attention mechanism, the hard partition strategy will destroy the spatial consistency of the large changed area that simultaneously exists within several adjacent patches. Thus, we design a CPB to preserve the spatial structures of the changed areas. Fig. 6 shows the architecture of the proposed CPB.

In Fig. 6, E_1 or E_2 is first convolved by two convolution layers with $2C$ filters. Then, we introduce two layers composed of dilated convolution filters to enlarge the receptive fields, whose efficiency has been validated by many methods, such as [54]. In the two dilated convolution (DCONV.) layers, $D = 3$ means that the dilation rate is set as 3. Through larger receptive fields,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

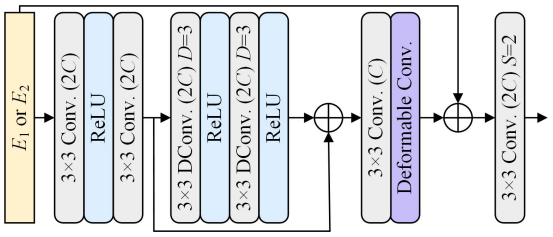


Fig. 6. Architecture of CPB.

the spatial information in adjacent patches can be learned more efficiently. In addition, considering the irregularity of the shape of changed areas, Zhu et al. [55] applied a deformable convolution layer to enhance the sensitivity for irregular contours or edges of changed areas. Then, we add the features containing spatial structural information into E_1 or E_2. Finally, a strided convolution layer is utilized to down-sample the feature to match the size of the features from other subnetworks. Compared to $F_{1,2}$ and $F_{2,2}$ extracted by 3×3 filters, $F_{1,1}$ and $F_{2,3}$ contains more spatial information at larger scales because of two DCONV. Layers are used in CPBs. For the 3×3 DCONV. with a dilation rate of 3, its receptive fields reach 9×9. Therefore, $F_{1,1}$ and $F_{1,2}$ are complementary in terms of spatial information. Finally, we directly introduce them into the concatenation stage and integrate them with $F_{1,3}$ and $F_{2,1}$ containing global information. Through the CPB, the spatial structures of the changed areas are preserved better, as it will be shown in the ablation study of Section IV.

D. Loss Function

After the proposed CTD-Former is constructed, it is trained to minimize the following cross-entropy loss

$$L_{\mathrm{CE}}\left(\tilde{G}, G\right) = -\left[G \odot \log \tilde{G} + (1 - G) \odot \log\left(1 - \tilde{G}\right)\right] \quad (13)$$

where $G \in \mathbb{R}^{H \times W}$ is the ground truth and $\tilde{G} \in \mathbb{R}^{H \times W}$ is the predicted change map. $\odot$ is the element-wise multiplication. In G, each element denotes the ground truth. When the element value is 1, the corresponding pixel is changed. So, for $1 - \tilde{G}$, the element with a value of 1 means that the pixel is unchanged. Cross-entropy measures the distance between G and $\tilde{G}$. The smaller the cross-entropy loss, the better the CD model. If the cross-entropy is 0, G and $\tilde{G}$ are matched perfectly.

IV. Experiments

In this section, we conduct comparison and ablation experiments on three CD datasets to demonstrate the effectiveness of the proposed CTD-Former. The datasets include LEVIR-CD [10], WHU-CD [56], and CLCD [20].

A. Description of Datasets

1) LEVIR-CD: This dataset is composed of 637 image pairs containing changes in buildings, which are collected from Google Earth images. These images consist of three bands: red (R), green (G), and blue (B). Their sizes are 1024×1024 and the spatial resolution is 0.5 m. For better training on GPUs, these images are segmented into patches with the size of 256×256. Then, the numbers of training/validation/test samples are 7120/1024/2048, respectively.

2) WHU-CD: In this dataset, one pair of aerial images is captured from Christchurch, New Zealand. The spatial size and resolution of the dataset are $32\,507 \times 15\,354$ and 0.2 m, respectively. R, G, and B bands are contained in the aerial image pair. The dataset is also divided into small patches with size 256×256 for training, validation, and test. In this article, training and validation datasets are composed of 6096 and 762 image pairs. All methods are tested on the remaining 762 image pairs.

3) CLCD: The CLCD dataset is constructed for the analysis of cropland change, which consists of 600 RGB image pairs with the size of 512×512. The resolutions of these images range from 0.5 to 2 m. In the article, these images are randomly grouped into three parts: 320/120/120 for training/validation/test of all methods, respectively.

B. Metrics

The performance of all methods on the above-mentioned datasets is evaluated by five metrics, including precision (P), recall (R), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA). The five metrics are computed as

$$\begin{aligned} P &= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \\ R &= \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \\ F1 &= 2PR/(P + R) \\ \mathrm{IoU} &= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN}) \\ \mathrm{OA} &= (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}) \end{aligned} \quad (14)$$

where TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively. For all metrics, larger values mean better performance.

C. Compared Methods and Implementation Details

To verify the effectiveness of the proposed method, seven methods are considered for comparison, including the FC-EF [6], the FC-Siam-conc [6], the FC-Siam-diff [6], the dual task constrained deep Siamese convolutional network (DTCDSCN) [57], the combination of Siamese network and NestedUNet (SNUNet) [58], the BIT [17], and the transformer-based CD (ChangeFormer) [21]. More details about these methods are summarized as follows.

1) *FC-EF:* Multitemporal images are directly concatenated and fed into an U-shaped fully convolutional neural network.
2) *FC-Siam-Diff:* Multitemporal images are encoded by a Siamese network. The differences in feature space are obtained for CD.
3) *FC-Siam-Conc:* The concatenation is implemented on the features of multitemporal images, which are extracted by a Siamese network.
4) *DTCDSCN:* Semantic segmentation of multitemporal images is used to facilitate CD between them. Channel attention and spatial attention are combined to extract more discriminative features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

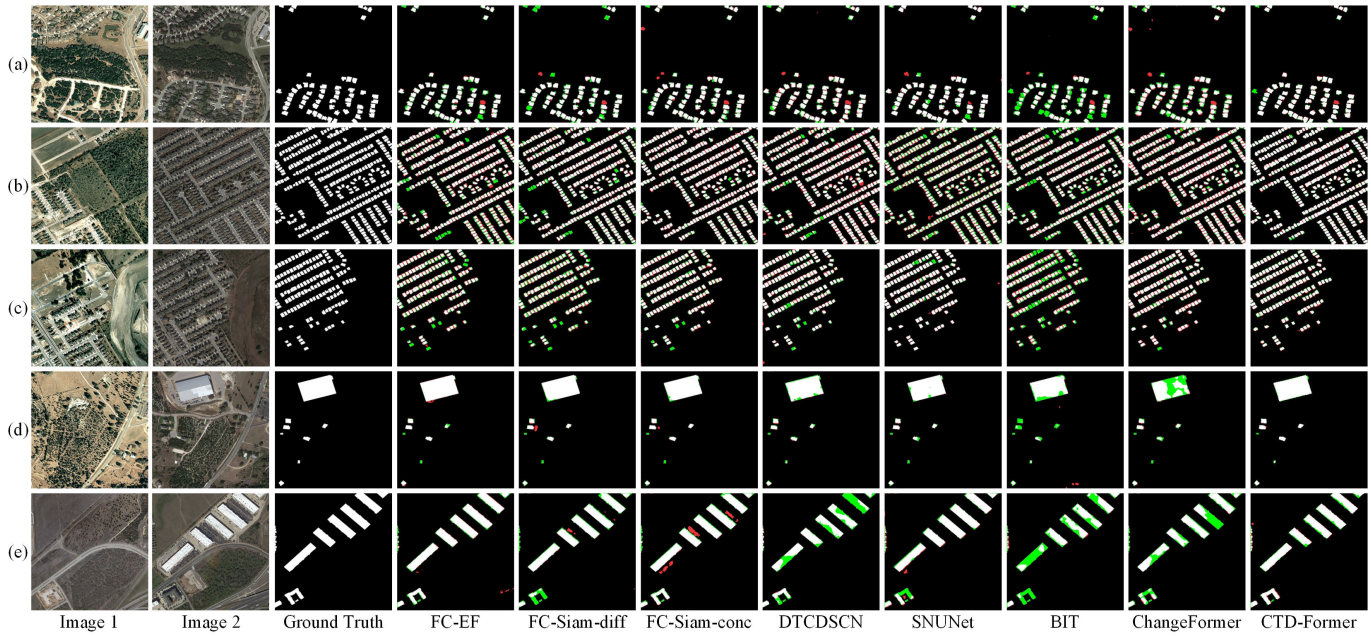


Fig. 7. Qualitative comparison of all methods on the LEVIR-CD dataset. (a)–(e) Prediction results of all methods on different image pairs.

5) *SNUNet:* The changed features are extracted by a SNUNet. Deep supervision is considered to constraint the intermediate features.
6) *ChangeFormer:* Features of multitemporal images are extracted by cascaded transformer blocks. The difference features are decoded by a lightweight MLP.
7) *BIT:* Tokenizers from multitemporal images are fused by a transformer encoder, whose output is projected by two transformer decoders to obtain the refined features.

The proposed CTD-Former is optimized by stochastic gradient descent (SGD). The initial learning rate is set as 0.01. The learning rate decreases by the multiplication of 0.1 every 400 epochs. The batch size is 6 for the training on LEVIR-CD and WHU-CD datasets. Because the size of images in the CLCD dataset is larger, we set the batch size as 2. When the proposed CTD-Former reaches 1000 epochs, the training is done. For all compared methods, the number of epochs is set as 1000 for a fair evaluation.

D. Comparison to State-of-the-Art Methods

1) Experiments on the LEVIR-CD Dataset: Fig. 7 shows the CD results of all methods on different image pairs. In Fig. 7, TP, TN, FP, and FN are highlighted by white, black, red, and green, respectively. From Fig. 7(a), we can see that medium-sized buildings are contained in the changed areas. Some changed buildings are not noticed by BIT and ChangeFormer and these areas are labeled in green. In the result of FC-Siam-conc, some unchanged areas are detected as the changed ones marked in red. In Fig. 7(b) and (c), there are many adjacent buildings with small sizes. For FC-Siam-conc, DTCDSCN, and ChangeFormer, the changed areas in their results tend to stick together owing to the FP in the results. In Fig. 7(d), DTCDSCN, BIT, and ChangeFormer cannot preserve the integrity of buildings with large sizes. As shown in Fig. 7, all changed areas are made up of the emerged buildings, which are similar in terms of spatial and spectral information. So, similar relationships in these areas can be easily found and the relation changes of objects in multitemporal images are modeled more efficiently. Then, the changed areas are detected better by the proposed CTD-Former.

TABLE I
QUANTITATIVE RESULTS OF ALL METHODS ON THE LEVIR-CD DATASET

Metric (%)	P	R	F1	IoU	OA
FC-EF	91.15	89.57	90.34	83.54	98.17
FC-Siam-diff	92.93	87.02	89.73	82.67	98.15
FC-Siam-conc	92.03	89.82	90.89	84.34	98.28
DTCDSCN	91.06	87.69	89.30	82.05	98.01
SNUNet	**93.83**	90.11	91.88	85.84	98.50
BIT	90.27	83.37	86.46	78.23	97.60
ChangeFormer	91.46	86.31	88.69	81.21	97.95
CTD-Former	93.60	**91.85**	**92.71**	**87.11**	**98.62**

Table I reports the quantitative results of all methods on the LEVIR-CD dataset. The best values are marked in bold. It can be found that the proposed CTD-Former achieves the highest values in terms of R, $F1$, IoU, and OA. The best P value is obtained by SNUNet. The largest P value means that the CD results of SNUNet contain fewer false positive areas. The performance of SNUNet may benefit from the dense connections in its backbone.

2) Experiments on the WHU-CD Dataset: Fig. 8 shows the results of all methods on the WHU-CD dataset. The results are presented with the same visualization used in Fig. 7. In the selected samples of Fig. 8, the changed areas are composed of buildings with large sizes. From Fig. 8(b), one can see that many FP areas exist in the results provided by DTCDSCN,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

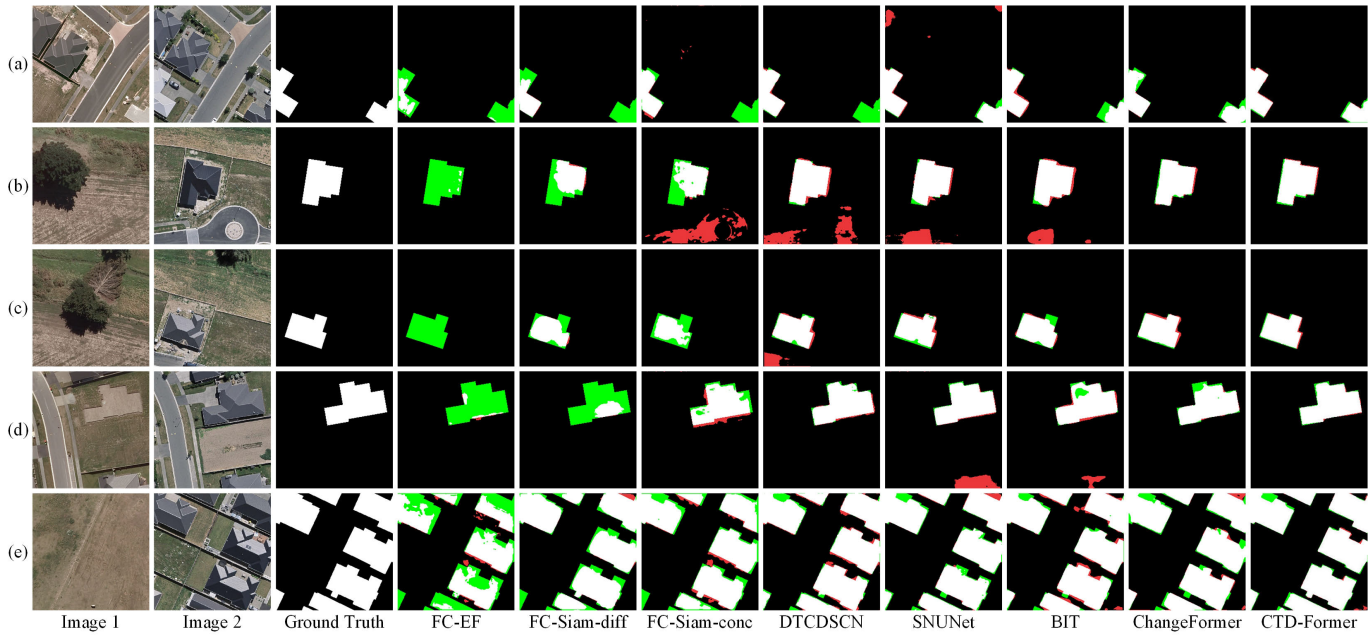


Fig. 8. Qualitative comparison of all methods on the WHU-CD dataset. (a)–(e) Prediction results of all methods on different image pairs.

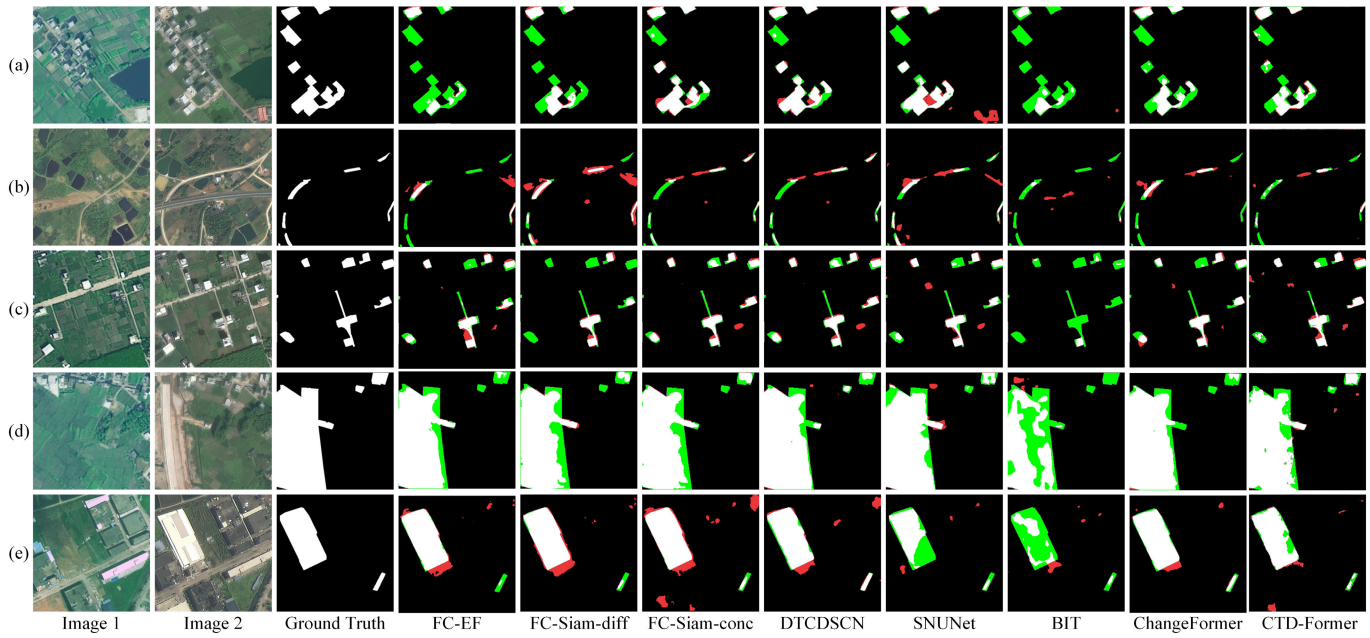


Fig. 9. Qualitative comparison of all methods on the CLCD dataset. (a)–(e) Prediction results of all methods on different image pairs.

SNUNet, BIT, and ChangeFormer. In the results of FC-EF and FC-Siam-diff, changed areas cannot be estimated efficiently. Similar to the LEVIR-CD dataset, the changed areas in the WHU-CD dataset are also composed of new buildings. Higher similarity among these buildings means that the relationships in changed areas are very different from those in unchanged areas. So, the proposed CTD-Former locates accurately the new buildings in the post-event image.

The numerical values on the WHU-CD dataset are listed in Table II. The proposed CTD-Former provides the best R, $F1$, IoU, and OA. The P value of our CTD-Former is also the best, which is followed by ChangeFormer. Thus, the overall performance of the proposed network is better than those of the compared methods.

3) Experiments on the CLCD Dataset: Fig. 9 shows the CD results of all methods on the CLCD dataset. The changes of objects in the CLCD dataset are more complicated than those in the first two datasets. The LEVIR dataset and WHU-CD dataset contain many obvious buildings, such as houses. Indeed, in the CLCD dataset, some changed areas are similar to the image backgrounds, which prevents them from being inferred accurately. For example, the changed areas with small sizes in Fig. 9(a)–(c) are mistaken as unchanged ones for some compared methods. To sum up, the proposed CTD-Former

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

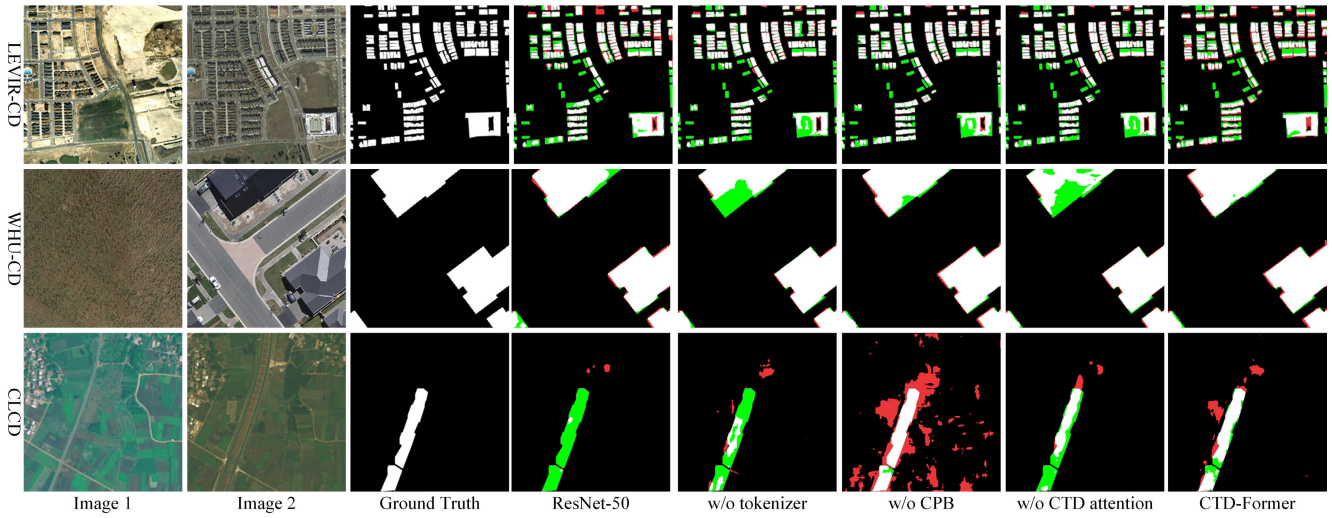


Fig. 10. Ablation study on the LEVIR-CD, the WHU-CD, and the CLCD datasets.

TABLE II
QUANTITATIVE RESULTS OF ALL METHODS ON THE WHU-CD DATASET

Metric (%)	P	R	F1	IoU	OA
FC-EF	92.61	83.26	87.30	79.40	98.10
FC-Siam-diff	94.52	89.32	91.74	85.65	98.70
FC-Siam-conc	93.94	92.83	93.37	88.19	98.90
DTCDSCN	93.24	94.39	93.80	88.86	98.95
SNUNet	92.61	83.26	87.30	79.40	98.10
BIT	87.92	93.41	90.46	83.72	98.29
ChangeFormer	96.28	92.95	94.55	90.08	99.12
CTD-Former	**96.74**	**97.03**	**96.89**	**94.11**	**99.50**

TABLE III
QUANTITATIVE RESULTS OF ALL METHODS ON THE CLCD DATASET

Metric (%)	P	R	F1	IoU	OA
FC-EF	82.83	76.20	79.06	69.19	94.81
FC-Siam-diff	83.13	74.60	78.10	68.19	94.73
FC-Siam-conc	81.65	79.94	80.77	70.98	94.84
DTCDSCN	85.54	81.75	83.51	74.28	95.70
SNUNet	85.92	82.62	84.19	75.11	95.84
BIT	81.42	70.27	74.39	64.51	94.17
ChangeFormer	84.41	81.36	82.80	73.40	95.47
CTD-Former	**87.29**	**83.17**	**85.08**	**76.24**	**96.12**

has a better performance in terms of changed buildings on the three datasets because the similarity in changed areas is higher and the relation changes in multitemporal images are extracted more efficiently.

Table III provides the quantitative results of all methods on the CLCD dataset. The best P and OA are from the proposed CTD-Former. Moreover, our method also achieves the best R, $F1$, and IoU values. This implies that the proposed method has a better overall performance.

E. Ablation Study

In this section, we verify the effectiveness of the modules used in the proposed CTD-Former on the three datasets. For example, ResNet-18 is substituted by ResNet-50 and tokenizer is replaced by an average-pooling layer. CPB is removed to analyze its influence on these datasets. Besides, we also replace the proposed CTD attention with vanilla attention. In the vanilla attention, the cross-temporal connections are removed and the query, key, and value are from the same image I_1 or I_2. Specifically, the vanilla attention is (2). In (2), only the correlations in single-temporal images are exploited and combined with the extracted features. The correlations among multitemporal images are ignored. Then, the proposed CTD attention is substituted by the vanilla attention.

We select some image pairs for comparison and their results are shown in Fig. 10. From the ablation results in Fig. 10, we can find that the ablation of CTD attention results in larger FN areas because the similarity among CD areas cannot be modeled accurately. Besides, instead of ResNet-18, ResNet-50 produces more FN areas although its FP area is small. When the tokenizer is ablated, more FN areas appear in the CD results. It may be because the semantic information cannot be modeled efficiently. Removing CPB destroys the integrity of changed areas and there are some irregular FP patterns. The complete CTD-Former can find the changed areas better than the compared other variants. In addition, Tables IV–VII give the evaluation metrics obtained in ablation studies. In Table IV, it can be found that the introduction of ResNet-50 cannot produce better CD performance. The reason for this may be that the number of parameters in ResNet-50 is much greater than that of ResNet-18, and ResNet-50 cannot be trained sufficiently on these datasets. Specifically, ResNet-50 contains 25.56M learnable parameters and the number of parameters in ResNet-18 is 11.69M. In Tables V–VII, we can see that the CD performance degrades when the modules adopted in the proposed network are replaced or removed.

F. Analysis of the CTD Attention

Considering the differentiability of (5) and (7), the element-wise square operation (Square) is adopted in the two equations to ensure the nonnegativity of elements of the matrix

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

COMPARISON OF RESNET-50 AND RESNET-18 ON THE LEVIR-CD, WHU-CD, AND CLCD DATASETS

Datasets	ResNet	P	R	F1	IoU	OA
LEVIR	ResNet-50	91.12	89.36	90.22	83.36	98.15
	ResNet-18	**93.60**	**91.85**	**92.71**	**87.11**	**98.62**
WHU-CD	ResNet-50	94.39	93.80	94.09	89.33	99.02
	ResNet-18	**96.74**	**97.03**	**96.89**	**94.11**	**99.50**
CLCD	ResNet-50	86.61	77.67	81.39	71.83	95.49
	ResNet-18	**87.29**	**83.17**	**85.08**	**76.24**	**96.12**

TABLE V

ABLATION STUDY OF THE PROPOSED CTD-FORMER ON THE LEVIR-CD DATASET

Tokenizer	CPB	CTD attention	P	R	F1	IoU	OA
×	✓	✓	93.68	91.18	92.39	86.61	98.57
✓	×	✓	93.63	91.33	92.44	86.70	98.58
✓	✓	×	**94.03**	90.68	92.28	86.45	98.56
✓	✓	✓	93.60	**91.85**	**92.71**	**87.11**	**98.62**

TABLE VI

ABLATION STUDY OF THE PROPOSED CTD-FORMER ON THE WHU-CD DATASET

Tokenizer	CPB	CTD attention	P	R	F1	IoU	OA
×	✓	✓	91.36	93.03	92.18	86.31	98.66
✓	×	✓	93.73	94.69	94.21	89.52	99.01
✓	✓	×	93.01	92.60	92.81	87.28	98.80
✓	✓	✓	**96.74**	**97.03**	**96.89**	**94.11**	**99.50**

TABLE VII

ABLATION STUDY OF THE PROPOSED CTD-FORMER ON THE CLCD DATASET

Tokenizer	CPB	CTD attention	P	R	F1	IoU	OA
×	✓	✓	81.87	76.54	78.90	69.00	94.67
✓	×	✓	79.19	78.19	78.95	68.93	94.25
✓	✓	×	81.34	76.60	78.73	68.80	94.59
✓	✓	✓	**87.29**	**83.17**	**85.08**	**76.24**	**96.12**

that records relation changes. As a common technique, the element-wise absolution value (Abs.) is also used to change the signs of elements. In this part, we investigate the influences of the operator on the three datasets. The numerical results are listed in Table VIII. The Table shows that the CD performance is improved by the Square operator. In particular, there are significant enhancements in the metric values on the WHU-CD dataset. The P value for this dataset is improved from 93.56% to 96.74%. The values of $F1$ and IoU are also boosted. For the CLCD dataset, the Square operator raises the value of P by 1.83%. $F1$ and IoU gain about 1%. The reason for this may be that the quadratic function further boosts the gap between values from unchanged and changed areas with respect to the Abs. operator. Thus, the discriminability of the features of changed areas is improved by the Square operation.

G. Analysis of the Tokenizer

This part presents the results of CTD-Former with different filter sizes. Tables IX–XI give the metric values obtained on the three datasets. When the size is small, the number of tokens

TABLE VIII

SQUARE VERSUS ABS. ON THE THREE DATASETS

Datasets	Operator	P	R	F1	IoU	OA
LEVIR-CD	Abs.	93.49	91.70	92.58	86.90	98.59
	Square	**93.60**	**91.85**	**92.71**	**87.11**	**98.62**
WHU-CD	Abs.	93.56	95.07	94.30	89.68	99.03
	Square	**96.74**	**97.03**	**96.89**	**94.11**	**99.50**
CLCD	Abs.	85.46	82.80	83.59	75.43	95.97
	Square	**87.29**	**83.17**	**85.08**	**76.24**	**96.12**

TABLE IX

INFLUENCE OF THE FILTER SIZE IN TOKENIZERS ON THE LEVIR-CD DATASET

Metric	Pre.	Rec.	F1	IoU	OA
4	**93.82**	91.53	92.64	87.00	98.61
8	93.55	91.54	92.52	86.82	98.59
16	93.60	**91.85**	**92.71**	**87.11**	**98.62**

TABLE X

INFLUENCE OF THE FILTER SIZE IN TOKENIZERS ON THE WHU-CD DATASET

Metric	Pre.	Rec.	F1	IoU	OA
4	93.90	94.30	95.09	90.98	99.22
8	93.52	92.56	93.04	87.65	98.89
16	**96.74**	**97.03**	**96.89**	**94.11**	**99.50**

TABLE XI

INFLUENCE OF THE FILTER SIZE IN TOKENIZERS ON THE CLCD DATASET

Metric	Pre.	Rec.	F1	IoU	OA
4	**87.29**	83.17	**85.08**	**76.24**	**96.12**
8	85.63	**84.29**	84.95	76.03	95.93
16	85.00	76.84	80.27	70.54	95.19

increases significantly, which results in high computational complexity. Besides, small filters can extract the refined details better, but they cannot capture the spatial information of the changed areas at larger scales efficiently. For convolution filters with large sizes, the complexity is reduced. But some changes at fine scales may be ignored because of the large receptive field. From Tables IX and X, one can see that the CD performance is improved with the increase of filter size for LEVIR-CD and WHU-CD datasets. However, the best performance on the CLCD dataset is produced when we set the size of filters in tokenizers as 4×4. The reason for this may be that the CLCD dataset contains smaller and thin changed areas as shown in Fig. 9. These changed areas in the CLCD dataset have to be modeled by filter with small sizes. Finally, the size of convolution filters in the tokenizer of CTD-Former is set as 16×16 for LEVIR-CD and WHU-CD datasets. For the CLCD dataset, the size is 4×4.

H. Training Convergence

Fig. 11 shows the $F1$-score versus the number of epochs for the three datasets. We can see that these values have similar growth trends on all datasets. For the WHU-CD dataset, the $F1$ value in the training stage approaches 99%. The $F1$ values on LEVIR-CD and CLCD datasets are also larger than 90%.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

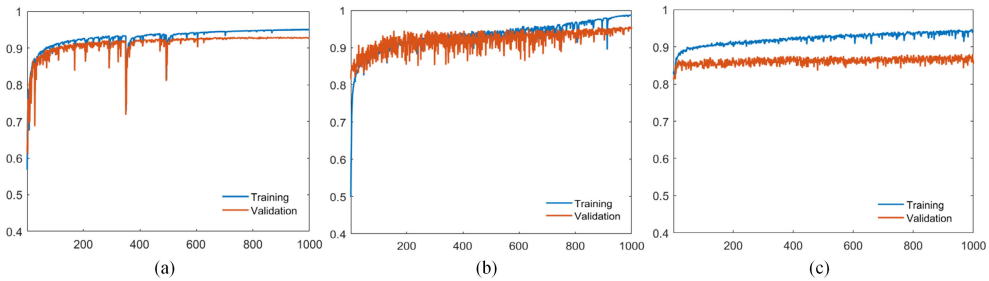


Fig. 11. $F1$-score of models for each training epoch. (a) LEVIR-CD dataset. (b) WHU-CD dataset. (c) CLCD dataset.

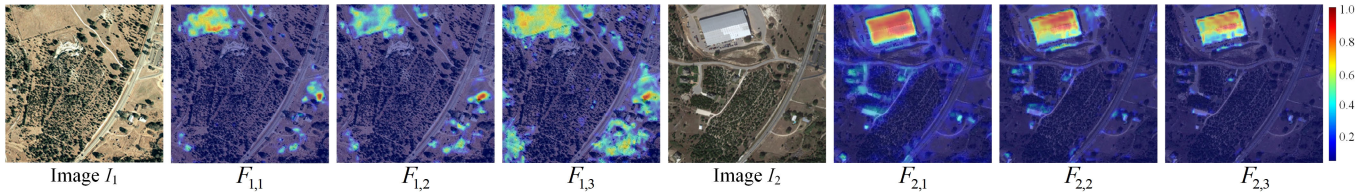


Fig. 12. Feature visualization on the LEVIR-CD dataset.

TABLE XII
TIME COMPARISON ON THE THREE DATASETS

Methods	LEVIR-CD		WHU-CD		CLCD	
	Training(h)	Test(s)	Training(h)	Test(s)	Training(h)	Test(s)
FC-EF	182.45	108.48	156.21	40.38	8.20	6.36
FC-Siam-Diff	195.80	107.38	167.64	39.97	8.80	6.29
FC-Siam-Conc	209.15	109.29	179.07	40.68	9.40	6.41
DTCDSCN	213.60	214.77	182.88	79.94	9.60	12.59
SNUNet	246.98	142.75	211.46	53.13	11.10	8.37
BIT	347.10	163.16	297.18	60.73	15.60	9.56
ChangeFormer	396.05	126.79	339.09	47.19	17.80	7.43
CTD-Former	404.95	187.43	346.71	69.76	18.20	10.99

The values of different metrics converge when the number of epochs reaches 1000. For WHU-CD and CLCD datasets, the variances of $F1$ at the validation stage are larger than that of the LEVIR-CD dataset, probably because the changed areas of the two datasets are more complex and difficult to estimate. In addition, there is no significant growth in terms of validation performance when the number of epochs is close to 1000. So, the maximum number of epochs is set as 1000.

I. Feature Visualization

To better understand the feature learning in the proposed CTD-Former, we take one image pair from the LEVIR-CD dataset as an example and employ Grad-CAM [59] to visualize the features that are fed into the concatenation module in Fig. 12. From the figure, we can see that the CD areas can be captured by these features. For the features of image I_1, the changed areas can be located although the buildings are not in the image I_1. Some weights appear in the unchanged areas of image I_1, which may be caused by the salient objects in these areas. In the feature $F_{2,3}$ of image I_2, the newly emerged buildings are found more efficiently, which verifies the effectiveness of the proposed CTD attention. Besides, the large weights in the features of image I_2 suppress weights in the unchanged areas of image I_1, and the changed areas are inferred efficiently.

J. Time Comparison

This part gives the training and testing time of all methods. In the experimental section, all methods are trained and tested on a server with an Intel Core i7-9700 processor, 3.0 GHz, an NVIDIA 2080Ti GPU, and 11-GB memory. In Table XII, the number of epochs of all DNN-based methods is set as 1000 for a fair comparison. Besides, Table XII reports the total test time of all images in test datasets. From Table XII, we can see that the training time of the proposed CTD-Former is considerable. The reason for this may be that matrix multiplication is performed twice in (5) and (7). The increase in training time is mainly caused by the estimation of relation changes. For ChangeFormer and BIT, matrix multiplication is implemented only once for the calculation of self-attention, because these methods ignore the relation changes of objects in multitemporal remote sensing images. Besides, FC-EF, FC-Siam-Diff and FC-Siam-Conc have better performance in training and test time, but their CD performance is limited. In Table XII, it also can be observed that the proposed CTD-Former is very close to ChangeFormer and BIT in terms of the test time.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this article, we proposed the CTD-Former for CD in remote sensing images. The proposed CTD-Former embeds the relation changes into features from multitemporal images through the CTD attention. Because the differences in terms of the relations of objects are recorded in the CTD attention, the changed areas are better distinguished from the complex unchanged areas. In CTD-transformer encoders, the features are extracted from tokens of images at the coarse scale. To enhance the details of these features, CTD-transformer decoders are used to achieve the up-sampling of them. In addition, CPBs are also designed to complement the features extracted by CTD-transformer. Experiments on three benchmark datasets verify the effectiveness of the proposed CTD-Former.

References

[1] F. Bovolo and L. Bruzzone, "The time variable in data fusion: A change detection perspective," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 8–26, Sep. 2015.

[2] Y. Guan, Y. Zhou, B. He, X. Liu, H. Zhang, and S. Feng, "Improving land cover change detection and classification with BRDF correction and spatial feature extraction using Landsat time series: A case of urbanization in Tianjin, China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4166–4177, 2020.

[3] F. Bovolo and L. Bruzzone, "A split-based approach to unsupervised change detection in large-size multitemporal images: Application to tsunami-damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1658–1670, Jun. 2007.

[4] D. Pirrone, F. Bovolo, and L. Bruzzone, "An approach to unsupervised detection of fully and partially destroyed buildings in multitemporal VHR SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5938–5953, 2020.

[5] D. Wen et al., "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 68–101, Dec. 2021.

[6] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[7] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[8] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603216.

[9] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.

[10] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[11] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[12] M. Wang, K. Tan, X. Jia, X. Wang, and Y. Chen, "A deep Siamese network with hybrid convolutional feature extraction module for change detection based on multi-sensor remote sensing images," *Remote Sens.*, vol. 12, no. 2, p. 205, Jan. 2020.

[13] Z. Dai, B. Cai, Y. Lin, and J. Chen, "UP-DETR: Unsupervised pre-training for object detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1601–1610.

[14] F. Zhang, K. Zhang, and J. Sun, "Multiscale spatial–spectral interaction transformer for pan-sharpening," *Remote Sens.*, vol. 14, no. 7, p. 1736, Apr. 2022.

[15] K. Zhang et al., "Panchromatic and multispectral image fusion for remote sensing and Earth observation: Concepts, taxonomy, literature review, evaluation methodologies and challenges ahead," *Inf. Fusion*, vol. 93, pp. 227–242, May 2023.

[16] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.

[17] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[18] F. Song, S. Zhang, T. Lei, Y. Song, and Z. Peng, "MSTDSNet-CD: Multiscale Swin transformer and deeply supervised network for change detection of the fast-growing urban regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6508505.

[19] N. Shi, K. Chen, and G. Zhou, "A divided spatial and temporal context network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4897–4908, 2022.

[20] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[21] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 207–210.

[22] M. Zaheer et al., "Big Bird: Transformers for longer sequences," in *Proc. 34th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 17283–17297.

[23] C. Yun, Y.-W. Chang, S. Bhojanapalli, A. S. Rawat, S. Reddi, and S. Kumar, "$O(n)$ connections are expressive enough: Universal approximability of sparse transformers," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 13783–13794.

[24] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[25] S. Marchesi and L. Bruzzone, "ICA and kernel ICA for change detection in multispectral remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2009, pp. 980–983.

[26] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, Apr. 1998.

[27] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi-and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[28] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[29] F. Bovolo, "A multilevel parcel-based approach to change detection in very high resolution multitemporal images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 1, pp. 33–37, Jan. 2009.

[30] P. Lv, Y. Zhong, J. Zhao, and L. Zhang, "Unsupervised change detection based on hybrid conditional random field model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4002–4015, Jul. 2018.

[31] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009.

[32] G. Moser, E. Angiati, and S. B. Serpico, "Multiscale unsupervised change detection on optical images by Markov random fields and wavelets," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 725–729, Jul. 2011.

[33] M. Gong, P. Zhang, L. Su, and J. Liu, "Coupled dictionary learning for change detection from multisource data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7077–7091, Dec. 2016.

[34] W. Zhang, J. Li, F. Zhang, J. Sun, and K. Zhang, "Unsupervised change detection of multispectral images based on PCA and low-rank prior," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5001605.

[35] D. Xue, T. Lei, X. Jia, X. Wang, T. Chen, and A. K. Nandi, "Unsupervised change detection using multiscale and multiresolution Gaussian-mixture-model guided by saliency enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1796–1809, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.

[36] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[37] R. Liu, D. Jiang, L. Zhang, and Z. Zhang, "Deep depthwise separable convolutional network for change detection in optical aerial images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1109–1118, 2020.

[38] M. Han, R. Li, and C. Zhang, "LWCDNet: A lightweight fully convolution network for change detection in optical remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6507905.

[39] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[40] B. Fang, G. Chen, G. Ouyang, J. Chen, R. Kou, and L. Wang, "Content-invariant dual learning for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603317.

[41] F. Jiang, M. Gong, T. Zhan, and X. Fan, "A semisupervised GAN-based multiple change detection framework in multi-spectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 7, pp. 1223–1227, Jul. 2020.

[42] C. Ren, X. Wang, J. Gao, X. Zhou, and H. Chen, "Unsupervised change detection in satellite images with generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10047–10061, Dec. 2021.

[43] S. Saha, L. Mou, X. X. Zhu, F. Bovolo, and L. Bruzzone, "Semisupervised change detection using graph convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 607–611, Apr. 2021.

[44] X. Tang et al., "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609715.

[45] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[46] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

[47] K. Song and J. Jiang, "AGCDetNet: An attention-guided network for building change detection in high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4816–4831, 2021.

[48] J. Huang, Q. Shen, M. Wang, and M. Yang, "Multiple attention Siamese network for high-resolution image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406216.

[49] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[50] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[51] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[52] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[53] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[54] L. Ding, H. Tang, Y. Liu, Y. Shi, X. X. Zhu, and L. Bruzzone, "Adversarial shape learning for building extraction in VHR remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 678–690, 2022.

[55] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets V2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.

[56] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[57] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[58] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8007805.

[59] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



Kai Zhang (Member, IEEE) was born in Shanxi, China, in 1992. He received the B.S. degree in electrical engineering and automation from the North University of China, Taiyuan, China, in 2013, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2018.

He was a Post-Doctoral Fellow with the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento, Trento, Italy, from November 2021 to November 2022. He is currently an Associate Professor with the School of Information Science and Engineering, Shandong Normal University, Jinan, China. His research interests include multisource remote sensing image fusion, change detection, and deep learning.



Xue Zhao received the B.S. degree in computer science and technology from Qilu Normal University, Jinan, China, in 2021. She is currently pursuing the M.S. degree in computer science and technology with the School of Information Science and Engineering, Shandong Normal University, Jinan.

Her research interests include remote sensing image change detection and deep learning.



Feng Zhang received the B.S. degree in electronic information engineering from Shandong Normal University, Jinan, China, in 2012, and the M.S. degree in electronic information engineering from Xidian University, Xi'an, China, in 2016. She is currently pursuing the Ph.D. degree in computer science and technology with the School of Information Science and Engineering, Shandong Normal University.

Her main research interests are deep learning and image processing.



Lei Ding received the M.S. degree in photogrammetry and remote sensing from PLA Strategic Support Force Information Engineering University, Zhengzhou, China, in 2016, and the Ph.D. degree (cum laude) in communication and information technologies from the University of Trento, Trento, Italy, in 2021.

He is currently a Lecturer with PLA Strategic Support Force Information Engineering University. His research interests are related to semantic segmentation, change detection, and domain adaptation with deep learning techniques.

Dr. Ding is a Referee of many international journals, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.



Jiande Sun received the B.S. and Ph.D. degrees in communication and information system from Shandong University, Jinan, China, in 2000 and 2005, respectively.

From 2008 to 2009, he was a Visiting Researcher with the Institute of Telecommunications System, Technical University of Berlin, Berlin, Germany. From 2010 to 2012, he was a Post-Doctoral Researcher with the Institute of Digital Media, Peking University, Beijing, China, and also with the State Key Laboratory of Digital-Media Technology, Hisense Group, Qingdao, China. From 2014 to 2015, he was a DAAD Visiting Researcher with the Technical University of Berlin and the University of Konstanz, Konstanz, Germany. From 2015 to 2016, he was a Visiting Researcher with the School of Computer Science, Language Technology Institute, Carnegie Mellon University, Pittsburgh, PA, USA. He is currently a Professor with the School of Information Science and Engineering, Shandong Normal University, Jinan. He has published more than 60 journal articles and conference papers. He is a coauthor of two books. His research interests include multimedia content analysis, video hashing, gaze tracking, image/video watermarking, and 2-D-to-3-D conversion.



Lorenzo Bruzzone (Fellow, IEEE) received the Laurea (M.S.) degree (summa cum laude) in electronic engineering and the Ph.D. degree in telecommunications from the University of Genoa, Genoa, Italy, in 1993 and 1998, respectively.

He is currently a Full Professor of telecommunications with the University of Trento, Trento, Italy, where he teaches remote sensing, radar, and digital communications. He is also the Founder and the Director of the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento. He is also the Principal Investigator of many research projects. Among the others, he is the Principal Investigator of the Radar for icy Moon Exploration (RIME) Instrument in the framework of the JUpiter ICy moons Explorer (JUICE) Mission of the European Space Agency (ESA) and the Science Lead of the High-Resolution Land Cover Project in the framework of the Climate Change Initiative of ESA. His papers are highly cited, as proven from the total number of citations (more than 37 000) and the value of the H-index (89). His research interests are in the areas of remote sensing, radar and synthetic aperture radar (SAR), signal processing, machine learning, and pattern recognition. He promotes and supervises research on these topics within the frameworks of many national and international projects.

Dr. Bruzzone has ranked the First Place in the Student Prize Paper Competition of the 1998 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Seattle, WA, USA, in July 1998. He was a recipient of many international and national honors and awards, including the recent IEEE GRSS 2015 Outstanding Service Award, the 2017 and 2018 IEEE IGARSS Symposium Prize Paper Awards, and the 2019 WHISPER Outstanding Paper Award. He was invited as a keynote speaker of more than 40 international conferences and workshops. Since 2003, he has been the Chair of the SPIE Conference on Image and Signal Processing for Remote Sensing. He was a Guest Coeditor of many special issues of international journals. He is the Co-Founder of the IEEE International Workshop on the Analysis of Multitemporal Remote-Sensing Images (MultiTemp) series and is a member of the Permanent Steering Committee of this series of workshops. Since 2009, he has been a member of the Administrative Committee of the IEEE Geoscience and Remote Sensing Society (GRSS), where he has been the Vice-President for Professional Activities since 2019. He has also been the Founder of *IEEE Geoscience and Remote Sensing Magazine*, for which he has been the Editor-in-Chief from 2013 to 2017. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He has been a Distinguished Speaker of the IEEE Geoscience and Remote Sensing Society from 2012 to 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:56:32 UTC from IEEE Xplore. Restrictions apply.